



Recorded office sensors

Categorical shared exclusions, signed contributions and continuous support

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Evidence status: the Rust example has compiled, passed Clippy, built in release mode and evaluated all three recordings. It produced categorical MGW results and the three expected continuous-support refusals. A separate calculation from integer event counts agrees with all 21 compared MI and net-atom values within $1e-12$ nats. These are finite numerical checks, not a formal Rust refinement or population-calibration theorem. PDF layout and reproduction are assessed separately from these numerical checks.

This example uses recorded room measurements and photograph-derived occupancy labels. Four training-fitted bins for light and CO₂ give a small categorical object that the existing shared-exclusions API can evaluate. The same binary target prevents admission to the current continuous Ehrlich/Wibral estimator. Keeping both findings visible is part of the example.

A later comparison used the MGW source-matching union to construct a forecast. It had worse log loss and Brier score than light alone and joint matching on all three previously examined recordings. The follow-up section gives its definition, results and reproducible calculation. This case study demonstrates neither a PID-guided improvement nor an advantage over direct predictive comparisons.

Data and roles

UCI identifies this as experimental office-occupancy data, with occupancy labels obtained from photographs taken every minute. The download contains three time-series files. UCI lists temperature in °C, relative humidity in %, light in lux, CO₂ in ppm, and humidity ratio in kg water vapor/kg air; it identifies humidity ratio as derived from temperature and relative humidity. These are the provider's provenance and units, not an independent audit of the instruments or labels. The retrieved record does not specify a timezone or the exact humidity-ratio formula. [UCI Occupancy Detection](#).

Role	File	Rows	Timestamp range, timezone unspecified	Occupied rows
Training	<code>datatraining.txt</code>	8,143	2015-02-04 17:51:00 → 2015-02-10 09:33:00	1,729
Earlier held-out	<code>datatest.txt</code>	2,665	2015-02-02 14:19:00 → 2015-02-04 10:43:00	972
Later held-out	<code>datatest2.txt</code>	9,752	2015-02-11 14:48:00 → 2015-02-18 09:19:00	2,049

The official first test recording predates training. It is a held-out earlier-recording comparison, not prospective evaluation. The second test recording follows training and provides the later-recording comparison. The recordings have no shared timestamps, but that does not establish statistical independence. No room, person, or building beyond the recorded office is held out here.





Recorded groups, explicit roles

Chronology matters. No random row split is used.

Earlier held-out

Feb 2–4 · 2,665 rows

Evaluated with later training maps

Training only

Feb 4–10 · 8,143 rows

Fit four bins per source

Later held-out

Feb 11–18 · 9,752 rows

Apply the fixed training maps

One CSV row: one timestamp, aligned sources and original target.

Light (lux) → bin 0–3

First categorical source

CO₂ (ppm) → bin 0–3

Second categorical source

Occupancy 0 / 1

Original same-row target

Nominal alphabet: $4 \times 4 \times 2$ states. Information: nats.

Current continuous API: binary target fails its support contract.

Source: UCI Occupancy Detection · All recordings are from February 2015.

Figure 1: Recording roles and aligned inputs. Only training rows determine the two source maps.

One original row supplies two simultaneous sources, light and CO₂, and the original binary target, occupancy. The target means the provider's unoccupied/occupied label; it does not encode head count, comfort, ventilation adequacy, energy use, a forecast horizon, or sensor quality. No lag, interpolation, resampling or timestamp join is used for this same-row task. An association with occupancy cannot identify its physical cause.

The table below transcribes the first raw data row in each file, including channels excluded from the main pair. The timestamps and decimal strings are preserved; the table is not a generated measurement.

Original field	Training row	Earlier test row	Later test row
Row identifier	1	140	1
Timestamp	2015-02-04 17:51:00	2015-02-02 14:19:00	2015-02-11 14:48:00
Temperature (°C)	23.18	23.7	21.76
Relative humidity (%)	27.272	26.272	31.1333333333333
Light (lux)	426	585.2	437.333333333333
CO ₂ (ppm)	721.25	749.2	1029.66666666667
Humidity ratio (kg water vapor/kg air)	0.00479298817650529	0.00476416302416414	0.00502101089021385
Occupancy (0/1)	1	1	1





What the files actually contain

Parsing the original CSV bytes gives 20,560 rows with no empty or nonfinite numeric cells, invalid target labels, duplicate timestamps or non-increasing timestamps. Every within-file interval is 59, 60 or 61 seconds. Counts of those intervals are 1,629/4,884/1,629 in training, 534/1,596/534 in the earlier test, and 1,951/5,849/1,951 in the later test. This supports approximate minute spacing, not an exact 60-second cadence. Missing physical events or label errors cannot be ruled out by complete CSV fields.

The header has seven named fields, while data rows have eight fields because a leading row identifier is unnamed. Identifiers restart across files and must not be used to join recordings. Timestamps are quoted in the first two files and unquoted in `datatest2.txt`. The example parses this specific, hash-checked format; it is not a general CSV loader.

The UCI table's integer type labels for temperature and light do not match the fractional values in the files. For example, training contains 7,653 fractional temperature readings and 1,085 fractional light readings. Decimal strings with long tails also occur. Printed precision, instrument resolution, within-minute aggregation and physical measurement accuracy are different things; the retrieved metadata does not establish the latter three.

Some complete five-channel tuples repeat: duplicate excess beyond one occurrence per distinct tuple is 867 in training, 82 in the earlier test and 488 in the later test. For the selected light/CO₂ pair the corresponding counts are 4,411, 643 and 4,493. Training has 889 distinct represented light values and 2,282 distinct represented CO₂ values. Such counts describe these records; they do not identify the population support or an effective independent sample size.

The following moments use the empirical distribution assigning mass $1/n$ to each row. Standard deviations divide by n . Bounds are observed minima and maxima, not population bounds. Light retains lux and CO₂ retains ppm; neither source is standardized for the categorical map.

Recording	Channel	Observed min	Observed max	Empirical mean	Empirical SD
Training	Light	0.000000	1546.333333	119.519375	194.743846
Training	CO2	412.750000	2028.500000	606.546243	314.301576
Earlier held-out	Light	0.000000	1697.250000	193.227556	250.163957
Earlier held-out	CO2	427.500000	1402.250000	717.906470	292.626801
Later held-out	Light	0.000000	1581.000000	123.067930	208.210599
Later held-out	CO2	484.666667	2076.500000	753.224832	297.080881

Full per-channel ranges, moments, decimal-place counts, smallest positive represented decimal spacings, repeated values, daily sample/label counts and raw sample rows are in [data facts](#). Daily groups are retained for interpretation; their sizes and occupancy proportions differ. February 7–8 in training and February 14–15 in the later test each contain 1,440 rows with an all-zero target. Under each of those daily empirical PMFs, target entropy and all Shannon mutual informations with that target are zero.





The fitted categorical object

Let L and C denote the light and CO_2 values in an aligned row, and Y the original occupancy bit. Fit four equal-width bins separately to L and C in `datatraining.txt`, then apply those same maps to all three recordings. Four bins and this source pair were fixed before viewing numerical outcomes; the tests do not select either choice.

The light edges, in lux, are $[0, 386.5833333333325, 773.166666666665, 1159.749999999975, 1546.33333333333]$. The CO_2 edges, in ppm, are $[412.75, 816.6875, 1220.625, 1624.5625, 2028.5]$. Interior edges go to the higher bin, the training minimum goes to bin 0, and the maximum goes to bin 3. Values below/above the training range go to the boundary bins under an explicitly declared `ClampToBoundary` policy. There is one above-range light value in each test and three above-range CO_2 values in the later test. No selected source falls below its training minimum. The default error policy would therefore reject affected held-out transforms; retaining those rows requires the explicit tail rule.

The empirical object in each recording is the joint PMF of $(Q_L(L), Q_C(C), Y)$, conditional on the retained training maps. Its nominal alphabet has $4 \times 4 \times 2 = 32$ states. Only 16, 10 and 16 joint states are observed in training, the earlier test and the later test, respectively. Six, two and four observed cells have fewer than five rows; minimum positive cell counts are one, three and one. An absent empirical cell is not evidence that the corresponding population probability is zero.

`EqualWidthQuantizer` retains each fitted map, its training/input identities and its occupancy diagnostics. The Rust example feeds the two transformed matrices and the unchanged binary target to `discrete_sxpid2_averaged_with_budget`, retaining both quantizer reports alongside the result. This explicit composition avoids quantizing an already binary target. The categorical call's encoding tag describes its supplied integer labels; the surrounding report identifies the fitted source estimand.

The PID quantity is the Makkeh–Gutknecht–Wibral shared-exclusions decomposition on that empirical PMF. It is measure-specific; Shannon MI and CMI alone do not determine its atoms. The reported atom net values are signed and must not be clamped. The implementation and provenance distinctions are documented in the [method catalog](#), with the defining paper, Makkeh–Gutknecht–Wibral, *Introducing a differentiable measure of pointwise shared information*, arXiv:2002.03356v5. The complete Rust output retains informative, misinformative and net components.

Shannon comparisons on the same PMF

Let $A = Q_L(L)$ and $B = Q_C(C)$ denote the fitted source labels. For finite variables X, Y in a nonempty recording of $N > 0$ rows, write $\hat{p}(x, y) = N_{x,y}/N$ for the fraction of rows in each joint state, and obtain marginal probabilities by summation. Entropy measures target uncertainty, while mutual information measures the average change in target information when the source is given:

$$H(Y) = - \sum_{y: \hat{p}(y) > 0} \hat{p}(y) \log \hat{p}(y), \quad I(X; Y) = \sum_{(x,y): \hat{p}(x,y) > 0} \hat{p}(x,y) \log \frac{\hat{p}(x,y)}{\hat{p}(x)\hat{p}(y)}.$$

Only positive joint masses enter the sum, so every displayed logarithm has a positive argument. For $X = (A, B)$, the same definition gives joint-source MI. Conditional MI averages the information within each supported $A = a$ group:

$$I(B; Y \mid A) = \sum_{(a,b,y): \hat{p}(a,b,y) > 0} \hat{p}(a,b,y) \log \frac{\hat{p}(a,b,y)\hat{p}(a)}{\hat{p}(a,b)\hat{p}(a,y)} = I(A, B; Y) - I(A; Y).$$





The last equality follows by expanding the two MI logarithms and cancelling their common marginal factors. Each formula describes one recording's empirical PMF. These are the classical Shannon quantities, separate from the measure-specific MGW redundancy defined below. Gray, *Entropy and Information Theory*, §5.5.

The following calculations were executed with standard-library counts and natural logarithms. They use each recording's labels and its empirical PMF, with the same training-fitted maps. They are descriptive plug-in values, not independent-sample estimates with calibrated uncertainty. For finite categorical Y , the population or empirical Shannon MI is bounded between zero and $H(Y)$. For this binary target, $H(Y)$ is at most $\ln 2 \approx 0.693147$ nats. CMI has its corresponding nonnegative conditional-entropy bound. These bounds follow from finite-target conditional entropy and the Shannon chain rule; see Gray, *Entropy and Information Theory*, corrected first edition (2023), §5.5, Lemmas 5.5.2, 5.5.6–5.5.7. They do not impose nonnegativity on shared-exclusions atoms.

Recording	$H(Y)$	$I(L;Y)$	$I(C;Y)$	$I(L,C;Y)$	$I(C;Y \text{ given } L)$	$I(L;Y \text{ given } C)$
Training	0.517027	0.464271	0.211048	0.469223	0.004952	0.258174
Earlier held-out	0.656090	0.569062	0.289596	0.577772	0.008710	0.288176
Later held-out	0.514103	0.481020	0.097547	0.482578	0.001558	0.385030

In this table L and C abbreviate their fitted categorical labels. CMI is computed by the chain-rule difference using the same empirical PMF, for example $I(C;Y \text{ given } L) = I(L,C;Y) - I(L;Y)$. CO_2 carries little additional plug-in information once the light bin is given, especially in the later recording. This is a statement about these maps and recordings, not an assertion that CO_2 has no useful role under different lighting, time alignment or prediction tasks. It is also not a PID redundancy or synergy estimate.

From one recorded row to the MGW decomposition

Write $A = Q_L(L)$ and $B = Q_C(C)$ for the fitted source labels. Choose a row uniformly from one recording of N rows. This defines its empirical PMF without an IID assumption about the physical time series. For a supported outcome (a, b, y) , let $E_{a,b}$ be the event $A = a$ **or** $B = b$. Let N_E , N_y and $N_{E,y}$ count rows in that event, rows with target y , and rows satisfying both conditions. The pointwise MGW redundancy, expressed with natural logarithms rather than the defining paper's base-2 logarithms, is

$$i_{\cap}^{\text{sx}}(y; a, b) = \log \frac{\hat{P}(Y = y \mid E_{a,b})}{\hat{P}(Y = y)} = \log \frac{N N_{E,y}}{N_E N_y}.$$

All three denominator/intersection counts are positive for an observed outcome. The disjunction is essential: it expresses the shared exclusion event in this definition. Replacing it by an intersection or a minimum of two mutual informations defines a different quantity. This is the published categorical MGW construction, applied to the empirical law; it is not a new estimator. Makkeh–Gutknecht–Wibral, arXiv:2002.03356v5.

For the first training row, light 426 lux maps to $a = 1$, CO_2 721.25 ppm maps to $b = 0$, and $y = 1$. There are $N = 8143$ rows, $N_E = 7922$ rows in the source disjunction, $N_y = 1729$ occupied rows, and $N_{E,y} = 1722$ occupied rows in the disjunction. Thus

$$i_{\cap}^{\text{sx}}(1; 1, 0) = \log \frac{8143 \cdot 1722}{7922 \cdot 1729} = \log \frac{14022246}{13697138}.$$





This is approximately 0.023458163 nats. The triple $(1, 0, 1)$ occurs in 510 rows; all of them have this same local value. To obtain averaged redundancy R , weight each supported triple's local value by its empirical frequency $N_{a,b,y}/N$:

$$R = \sum_{(a,b,y): N_{a,b,y} > 0} \frac{N_{a,b,y}}{N} i_{\cap}^{\text{sx}}(y; a, b).$$

The two-source reconstruction then gives the unique light contribution U_L , unique CO₂ contribution U_C , and synergy S :

$$\begin{aligned} U_L &= I(A; Y) - R, & U_C &= I(B; Y) - R, \\ S &= I(A, B; Y) - I(A; Y) - I(B; Y) + R. \end{aligned}$$

All terms in these equations use the same empirical PMF and natural logarithm. Averaging the separate informative and misinformative components gives the net values below. The complete output retains both components so a signed net value does not hide their difference.

Recording	Redundancy R	Unique light U_L	Unique CO ₂ U_C	Synergy S	Joint MI
Training	0.327008189	0.137262473	-0.115959738	0.120911606	0.469222531
Earlier held-out	0.440298062	0.128763882	-0.150701991	0.159411830	0.577771783
Later held-out	0.276238458	0.204781420	-0.178691053	0.180249065	0.482577890

Values are in nats and rounded for display. The largest absolute atom-sum residual in the unrounded Rust output is 2.22×10^{-16} nats. A separate direct calculation from integer disjunction counts reproduces the four net atoms and three MIs for each recording; its largest absolute difference from Rust is 1.67×10^{-16} nats. The [42 supported-state records](#) and [21 comparisons](#) expose each event count and local value. This finite check tests the defining redundancy as well as the reconstruction identities; it supplies no universal floating-point error bound.

Why a positive synergy is not a sensor-selection verdict

Subtracting the light-only identity from the joint identity gives

$$I(B; Y \mid A) = U_C + S.$$

For the later recording, the right-hand side is approximately $-0.178691053 + 0.180249065 = 0.001558012$ nats. The sizeable positive synergy is nearly cancelled by negative unique CO₂ information. Clamping the negative term would destroy this relation and greatly overstate the measured additional information.





Signed terms explain a small conditional MI

Later recording · fixed four-bin Light and CO₂ · binary occupancy

Unique CO₂

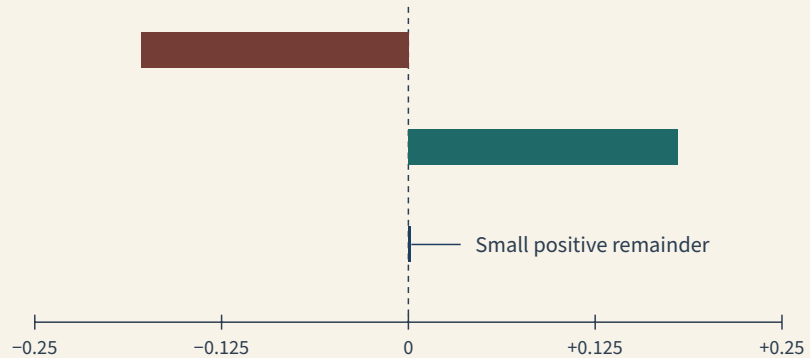
-0.178691053 nats

Synergy

+0.180249065 nats

Conditional MI

+0.001558012 nats



Unique CO₂ + synergy = I(CO₂ bin; occupancy | Light bin)

One common linear scale, in nats. Signed MGW contributions are not sensor fault scores.

Figure 2: Signed MGW contributions nearly cancel when computing the later recording's conditional mutual information.

Negative U_C is a signed contribution under this particular decomposition. It does not identify a faulty sensor, intentional misinformation, or a harmful causal effect. Conversely, positive S alone does not establish that installing CO₂ sensing improves a decision system. Here PID exposes how its signed contributions combine; CMI summarizes the additional empirical information, and the predictive scores below measure a separate fixed decision rule. A sensor recommendation would also require task loss, acquisition cost and an appropriate validation design.

Fixed predictive comparisons

All predictor probabilities are fitted on training only. The constant model uses the training occupancy fraction 1,729/8,143. For each one-source bin or pair of bins, the other models use the add-one Bernoulli estimate $(\text{occupied} + 1)/(\text{total} + 2)$. An unseen training cell gets probability 1/2. Classification declares occupied at probability at least 1/2. No held-out tuning or model selection was performed.

These are same-row occupancy classifiers. Accuracy, balanced accuracy and Brier score are dimensionless; log loss uses natural logarithms and is reported in nats per row. Accuracy and Brier score are bounded in $[0,1]$. Balanced accuracy averages the two class recalls and is undefined for a one-class daily group; the full held-out recordings both contain both classes. The retained daily JSON uses null for the undefined case. Smoothed probabilities and the nondegenerate training prior make every observed log loss finite. No population loss moment or calibration assumption is needed to calculate these finite empirical averages.





Recording	Predictor	Accuracy	Balanced accuracy	Brier score	Log loss
Earlier held-out	constant	0.635272	0.500000	0.254927	0.716812
Earlier held-out	light	0.969231	0.969649	0.024603	0.097773
Earlier held-out	co2	0.873546	0.855782	0.112753	0.389536
Earlier held-out	pair	0.978612	0.982509	0.022241	0.089744
Later held-out	constant	0.789889	0.500000	0.165969	0.514118
Later held-out	light	0.981952	0.966008	0.011556	0.047443
Later held-out	co2	0.782404	0.717718	0.164311	0.506560
Later held-out	pair	0.982363	0.966984	0.011683	0.050762

Higher accuracy is better; lower Brier score and log loss are better. On the later recording, adding CO₂ to the light-bin predictor corrects four classifications, but slightly worsens both probability scores. The CO₂-only classifier has lower overall accuracy than the constant classifier while having higher balanced accuracy. These comparisons show why a single favorable metric cannot establish a general benefit. No predictor was selected or trained using PID in these original comparisons, so none of these differences is attributed to PID. The subsequent OR-event development comparison below evaluates a separately specified MGW-event-inspired forecast.

Recording	Predictor	True negative	False positive	False negative	True positive	Rows in unseen training cells
Earlier held-out	constant	1693	0	972	0	0
Earlier held-out	light	1639	54	28	944	0
Earlier held-out	co2	1560	133	204	768	0
Earlier held-out	pair	1639	54	3	969	25
Later held-out	constant	7703	0	2049	0	0
Later held-out	light	7653	50	126	1923	0
Later held-out	co2	6388	1315	807	1242	0
Later held-out	pair	7653	50	122	1927	4

The 25 and four unseen pair-bin rows in the earlier and later tests matter: the fixed probability-1/2 and tie rule predicts occupancy for them. Improvements in thresholded classification are therefore not evidence of confident predictions for novel combinations. The exact counts, per-day evaluations and information values are retained in [descriptive comparisons](#).

Does the shared-exclusion event make a useful forecast?

A follow-up development comparison tested a specific interpretation of categorical Makkeh–Gutknecht–Wibral (MGW) redundancy. Its local definition uses the target probability within a source-matching **union**. Could that same event provide a useful simple forecast for the recorded light and CO₂ pair? This question concerns one predictor. It does not assume that maximizing redundancy or synergy improves a task.





The forecast and its assumptions

Keep the original training-fitted maps: four light bins, four CO₂ bins, and binary occupancy. For an observed source query (a, b) , select each training row whose light bin equals a **or** whose CO₂ bin equals b . A row that matches both is selected once. Both source values are used to specify the query; this is not a rule for an unavailable sensor.

Let $n_{a,y}$ count training rows with light bin a and target y , $n_{b,y}$ count rows with CO₂ bin b and target y , and $n_{ab,y}$ count rows matching both source bins and that target. Inclusion–exclusion gives the exact integer count

$$n_{\cup,y}(a, b) = n_{a,y} + n_{b,y} - n_{ab,y}, \quad y \in \{0, 1\}.$$

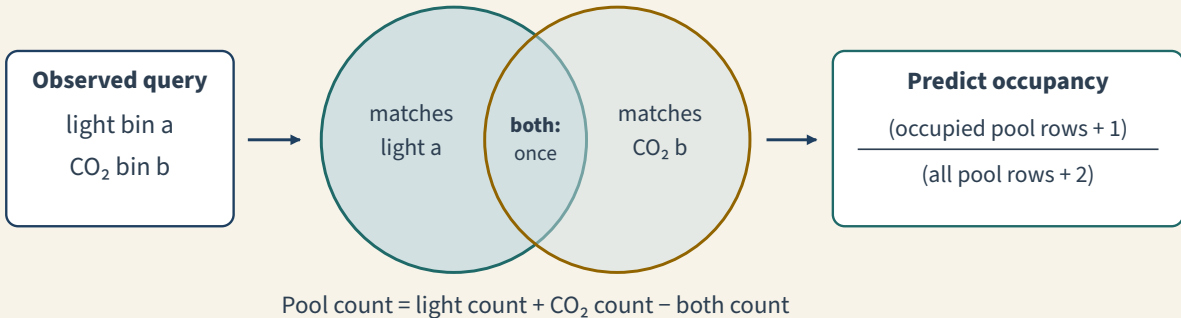
The subtraction removes the duplicate copy of every row in the intersection. Set $n_{\cup} = n_{\cup,0} + n_{\cup,1}$. Add one pseudocount for each target label **after** selecting the event:

$$q_{\cup}(y | a, b) = \frac{n_{\cup,y}(a, b) + 1}{n_{\cup}(a, b) + 2}.$$

The two numerators sum to the denominator, so the probabilities sum to one. Each is strictly between zero and one. An empty pool gives $(1/2, 1/2)$. Every OR pool in the training four-by-four query alphabet was nonempty; the empty-pool case was checked as a separate arithmetic control. Six joint-matching training cells were empty.

The comparison uses the same smoothing rule for constant, light-only, CO₂-only, joint matching, and OR matching forecasts. The original comparison's constant baseline instead used unsmoothed training prevalence. Its values are preserved and separately reproduced; the two constant forecasts are not equated. The other three baseline rules are unchanged.

From a complete query to a matching-row forecast



Both bins select the event. This is not a missing-sensor protocol.

Schematic sets; areas do not represent the observed counts. Author: Sepehr Mahmoudian.

Figure 3: The complete query selects a union of training rows. Matching both source values does not count a row twice.





What the connection to MGW does and does not say

For a finite law P of (A, B, Y) and a query (a, b) in its source support, define $E_{a,b} = \{A = a\} \cup \{B = b\}$ and $q_P(y | a, b) = P(Y = y | E_{a,b})$. At every supported triple (a, b, y) , the event contains that triple, so its probability and its target- y probability are positive. Using this report's natural-log (nat) convention instead of the defining paper's base-2 logarithms, the local MGW redundancy is

$$i_{\cap}^{\text{sx}}(y; a, b) = \log q_P(y | a, b) - \log P(Y = y).$$

Taking expectation under the **same** law P gives

$$R_P = \mathbb{E}_P[\log q_P(Y | A, B)] - \mathbb{E}_P[\log P(Y)] = H_P(Y) - L_P(q_P),$$

where $L_P(q) = \mathbb{E}_P[-\log q(Y | A, B)]$ is natural-log loss. This is a direct rewriting of the MGW definition, not a new redundancy measure or a novelty claim. The redundancy can be negative, so this identity alone does not guarantee an improvement over the marginal forecast. The query-specific events can overlap; selecting them uses the full source query. This identity alone supplies no fixed coarsened observation channel or missing-sensor protocol. It also does not identify a smoothed training forecast's loss on a different recording with that recording's MGW redundancy.

Results on the previously examined recordings

All forecasts were fitted on the 8,143 training rows. The 2,665-row earlier recording predates training; the 9,752-row later recording follows it. All three had already been examined before this follow-up was designed. These are development comparisons, not untouched confirmation. The calculation reused the published integer count tables; it did not parse the raw sensors, refit the quantizers, or rerun Rust.

Recording	Light log loss	Joint log loss	OR log loss	Light Brier	Joint Brier	OR Brier
Training	0.052883	0.048246	0.190148	0.011547	0.011361	0.050375
Earlier	0.097773	0.089744	0.222431	0.024603	0.022241	0.061581
Later	0.047443	0.050762	0.218677	0.011556	0.011683	0.061280

Lower values are better. Log loss is in nats per row; binary Brier score is the mean squared difference between the predicted occupancy probability and the recorded bit. OR matching loses to both light alone and joint matching on all three recordings under both scores. It beats the CO₂-only and smoothed-constant baselines. Full-precision results for all five rules are available as [CSV](#) and in the [complete calculation record](#), including exact forecast fractions, union witnesses and every specified check.

The retained reference run passed 244 specified checks, including exact integer union counts and rational probability normalization. Using each recording's own unsmoothed event posterior, $H_P(Y) - L_P(q_P)$ matched the retained Rust redundancy within 2.78×10^{-16} nats. A separate dense-count computation reproduced all five forecasts' metrics. It shared the data, formulas and Python logarithm implementation; this was implementation cross-checking, not independent field evidence.

The [portable evidence note](#) links the historical protocol and calculator, full results and alternate computation. A portable successor accepts an explicit input directory and verifies the two published input hashes. Its 24 September replay passed all 244 checks and reproduced every retained scientific result field exactly in the recorded arithmetic environment. This repeats fixed counts; it adds no new observations. The historical calculator retains its expired deadline as executed source, not as a current rerun command.





Implications

This result rejects this OR-matching forecast as an improvement over the stated light and joint baselines on these records. Direct task loss already reaches that conclusion. The experiment does not test a PID-guided model intervention, and it does not refute other uses of MGW. It prevents the event-posterior identity from being treated as an automatic fusion advantage. A later positive claim needs a specific mechanism and comparison on new independent recording units, with representation and tuning choices fixed before evaluation.

Rust implementation and computational cost

The existing Rust example computes empirical MGW PID2; it does not expose this new forecast as a production API. This comparison is a Python publication-reference calculation on a $4 \times 4 \times 2$ count table. Let K_A, K_B be the source alphabet sizes and K the number of occupied training states. The reference scans those K states for each of $K_A K_B$ source queries, so forecast construction uses $O(K_A K_B K)$ count operations for a fixed number of models. It does not reread N raw rows. A marginal-table implementation can instead count the rows in $O(N)$ work and form all OR counts by inclusion-exclusion in $O(K_A K_B)$ work; dense storage then uses $O(K_A K_B)$ entries. Scoring a fixed number of forecasts costs $O(K_e)$ for a recording with K_e occupied states. The separate same-law identity check rescans those states for each supported query, so its present implementation costs $O(K_e^2)$; it is not part of forecast lookup. Either precomputed forecast uses one lookup per query. These are arithmetic-operation counts; integer bit costs depend on count size. There is no production Rust forecast or comparative runtime benchmark. Feature extraction, raw-data parsing and quantizer fitting are separate costs. No Gaussian, continuous-density or hyperbolic model is assumed. Temporal dependence does not prevent these empirical sums, but a population confidence statement needs a justified sampling model; none is supplied.

The defining source is Makkeh, Gutknecht and Wibral, *Introducing a differentiable measure of pointwise shared information*, arXiv:2002.03356v5. The measurements and labels are from UCI Occupancy Detection. This follow-up forecast specification, calculation and scoped negative disposition are project work by Sepehr Mahmoudian. Cite this report with its exact repository revision.

The continuous Ehrlich/Wibral track

The continuous shared-exclusions method of Ehrlich, Schick-Poland, Makkeh, Lanfermann, Wollstadt and Wibral is a different functional and estimator from the fitted categorical calculation. Its primary record describes a continuous formulation and nearest-neighbor estimator; its reported Honda application is a **simulated** energy-management system. That example does not provide recorded office measurements. Ehrlich et al., *Partial Information Decomposition for Continuous Variables based on Shared Exclusions: Analytical Formulation and Estimation*, arXiv:2311.06373v3.

The current repository implementation is experimental and restricted to equal ambient source dimensions, declared regular full-dimensional population support and a specified source-coordinate gauge. The two scalar sources meet the dimension count, but the binary occupancy target fails the required full-dimensional continuous joint-law condition. With the optional feature enabled, the example called the complete continuous PID2 report API with `KnownAtomicOrMixed`. All three calls returned the required `UnsupportedSupportContract` variant and no numerical continuous PID. This tests those three declared-support refusals; it does not prove all support guards. The retained raw source units are lux and ppm, without jitter or scaling; those coordinate choices would matter for a continuous shared-exclusions estimand.





Repeated recorded values and temporal sampling are separate issues. An underlying analog quantity may be continuous while its stored readings are rounded or aggregated. Finite observations cannot establish a population density, finite population MI, IID rows, regular neighbor geometry or estimator calibration. Uncorrelated observations need not be independent; striding the recordings does not prove IID. Adding artificial noise would define a new observation model and target, not repair the original occupancy estimand.

Humidity ratio is especially instructive. UCI calls it derived from temperature and relative humidity, so it must not be advertised as another independent instrument. Yet the CSV contains three represented temperature/humidity pairs with slightly different humidity-ratio strings. For example (21.7 °C, 20.075%) occurs at training row IDs 2,584 and 2,585 with humidity ratios 0.00321644884277157 and 0.00321644884277158. The [three exact witness pairs](#) preserve that negative finding. It prevents claiming an exact binary64 functional relation from the printed inputs. The physical derivation formula, input precision and pressure assumptions remain unverified; the tiny discrepancies do not establish a new independent noise source.

For standard Borel random variables X and Y with measurable f and $Y = f(X)$ almost surely, a non-atomic target law implies infinite MI. A countably supported target instead has MI equal to its entropy (which may itself be infinite); a binary target has entropy at most $\ln 2$, and a constant target has MI zero. Failure of full-dimensional continuous support alone is insufficient to infer infinite MI. These are consequences of the general-alphabet information definition and finite quantization; see [Gray, §5.5, Lemma 5.5.1 and Corollary 5.5.6](#). The empirical constant-target daily groups above are a concrete finite example of the zero-MI case.

A separate continuous question could use sources `temperature(t)` and `light(t)` with target `CO2(t + 15 minutes)`. It would concern future CO₂, not occupancy. A separately logged standard-library alignment calculation used exact timestamp matches within each recording. It retained only 3,793 training, 1,237 earlier-test and 4,545 later-test rows. Fifteen rows at the end of each recording have no in-record future target; a further 4,335, 1,413 and 5,192 internal rows, respectively, have no exact 900-second match. These counts are retained in the [alignment evidence](#). The observed near-minute cadence does not justify silently replacing an exact 15-minute lag with a 15-row shift. A declared alignment or rounding rule must retain its exclusions and must not cross recording boundaries. Before any continuous number is admitted, the population observation model, treatment of rounded values, dependence model, target finiteness conditions and training-fitted source gauge must be stated and defended. Their truth is not established here. Only the alignment description has been executed; continuous estimation on this changed target remains conditional and unexecuted.

Execution, resource bounds and retained failures

The executed example checked the three original file hashes and row counts, fitted on training only, and emitted complete quantizer and categorical reports. Rust's source edges, tail counts and Shannon values agree with the retained descriptive calculations. Each result also includes the library's conservative resource estimate. The declared per-call budget is 64 MiB, one thread, 1,000,000 pairwise distances and a 4,000,000,000-operation hint. These ceilings do not guarantee process RSS or elapsed time.

Recording	Estimated allocation exposure (bytes)	Operation hint	Pairwise distances
Training	7,366,072	2,127,384,267	0
Earlier held-out	2,413,960	229,052,508	0
Later held-out	8,820,608	3,049,939,088	0





The operation hint assumes that every row may occupy a different joint state. The actual categorical calculation iterates over the observed empirical states. These resource-accounting units are not CPU-instruction counts, and the allocation estimate is not a measured peak RSS. The full output is written after all three recordings succeed; a failed call instead leaves an error and nonzero exit.

Three unsuccessful execution steps are retained in the development record. First, formatting a SHA-256 array directly as hexadecimal failed to compile; explicit byte formatting fixed that defect without changing the expected hashes. Second, mixed Rust compiler versions caused Clippy to reject compiled dependencies; a consistent Rust 1.96 toolchain and separate build directory resolved that environment mismatch. Third, the original 100,000,000-operation budget was too small: the first categorical call requested 2,127,384,267 and correctly refused to run. The explicit 4,000,000,000 ceiling covers all three reported estimates; the library guard was not changed.

The final default check, experimental Clippy check, release build and example process all exited successfully. The example process completed in one observed supervised interval of about 0.37 seconds under a 60-second limit, 2 MiB per-stream limits and a sampled 4 GiB process-group RSS ceiling. This single run is not a throughput benchmark or hard operating-system resource guarantee.

No bootstrap interval, permutation null, significance test or population estimator guarantee follows from this run. Uncertainty analysis would need a justified sampling design that accounts for the time series. PDF rendering, source agreement and numerical verification are separate checks.

Attribution and source identity

Data and transcribed rows: Luis Candanedo (2016), *Occupancy Detection*, UCI Machine Learning Repository, DOI 10.24432/C5X01N, made available by UCI under CC BY 4.0. Transformations here comprise four training-fitted source bins, explicit boundary clamping, empirical summaries and fixed train-only conditional-frequency baselines. The associated paper by Candanedo and Feldheim is *Accurate occupancy detection of an office room from light, temperature, humidity and CO₂ measurements using statistical learning models*, *Energy and Buildings* 112, 28–39, institutional primary record, DOI 10.1016/j.enbuild.2015.11.071. That record dates online publication to 2 December 2015; UCI's data donation is in 2016. No access-restricted full paper or photographs were retrieved.

The official ZIP is 335,713 bytes with SHA-256 4ae3f46aa98eedff564a9f6924d1635173e2fd2c816004342a9be93076d3a81a. Its three members total 1,497,104 uncompressed bytes. Individual hashes are retained in the example and data-facts JSON. Hashes identify the retrieved bytes; they do not certify measurements, labels, license authority or scientific validity.

Suggested citation: Sepehr Mahmoudian (2026), *Recorded office sensors: what categorical and continuous shared exclusions can establish*. Include the exact repository commit or release. For the software, use CITATION.cff; the method and dataset citations above identify their separate authorship.

